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25K-0899

BCE-2C

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Q No 1

(a) $\iint_D (x+2y) dA$, $y=2x^2$ and $y=1+x^2$

$$2x^2 = 1+x^2$$

$$x^2 = 1 \rightarrow (x = \pm 1)$$

so, limits

$$2x^2 \leq y \leq 1+x^2 \quad -1 \leq x \leq 1$$

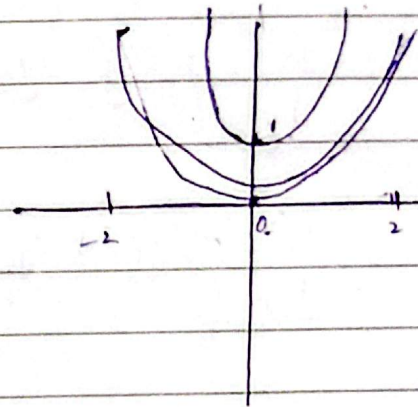
$$= \int_{-1}^1 \int_{2x^2}^{1+x^2} (x+2y) dy dx$$

$$= \int_{-1}^1 \left[xy + y^2 \right]_{2x^2}^{1+x^2} dx = \int_{-1}^1 x + x^3 + (1+x^2)^2 - (2x^3 + 4x^4) dx$$

$$= \int_{-1}^1 (x - x^3 + 2x^2 - 3x^4) dx = \left[\frac{x^2}{2} - \frac{x^4}{4} + \frac{2x^3}{3} - \frac{3x^5}{5} \right]_{-1}^1$$

$$= \left(\frac{1}{2} - \frac{1}{4} + \frac{2}{3} - \frac{3}{5} \right) - \left(\frac{1}{2} - \frac{1}{4} - \frac{2}{3} + \frac{3}{5} \right)$$

$$= \frac{32}{15}$$



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(b) $z = x^2 + y^2$

$$V = \iint_D x^2 + y^2 \, dA$$

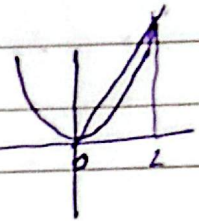
$$y = 2x \text{ \& \& } y = x^2$$

$$x^2 = 2x$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$x = 0, 2$$



$$V = \int_0^2 \int_{x^2}^{2x} x^2 + y^2 \, dy \, dx \quad x^2 \leq y \leq 2x, \quad 0 \leq x \leq 2$$

$$= \int_0^2 \left[x^2 y + \frac{y^3}{3} \right]_{x^2}^{2x} dx = 2x^3 + \frac{8x^3}{3} - x^4 - \frac{x^6}{3}$$

$$= \int_0^2 \left(\frac{14x^3}{3} - x^4 - \frac{x^6}{3} \right) dx = \left[\frac{14}{12} x^4 - \frac{x^5}{5} - \frac{x^7}{21} \right]_0^2$$

$$= \frac{14}{12} (2)^4 - \frac{(2)^5}{5} - \frac{2^7}{21} - 0 = \boxed{\frac{216}{35}}$$

Qno 2

$$\iiint_R xyz^2 \, dV \quad R = \{ (x, y, z) \mid 0 \leq x \leq 1, -1 \leq y \leq 2, 0 \leq z \leq 3 \}$$

By property

$$= \int_0^1 x \, dx \int_{-1}^2 y \, dy \int_0^3 z^2 \, dz$$

$$= \left[\frac{x^2}{2} \right]_0^1 \left[\frac{y^2}{2} \right]_{-1}^2 \left[\frac{z^3}{3} \right]_0^3$$

$$= \left(\frac{1}{2} \right) \left(2 - \frac{1}{2} \right) (9) = \boxed{\frac{27}{4}}$$

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$$(b) \iiint \frac{1}{x^3} dv \quad 0 \leq y \leq 1, 0 \leq z \leq y^2$$

$$= \int_0^1 \int_0^{y^2} \int_1^{z+1} \frac{1}{x^3} dx dz dy = \left[-\frac{x^{-2}}{2} \right]_1^{z+1}$$

$$= -\frac{(z+1)^{-2}}{2} + \frac{1}{2} = -\frac{(z+1)^{-2}}{2} + \frac{1}{2} - \frac{1}{2(z+1)^2}$$

$$= \int_0^1 \int_0^{y^2} \left[\frac{1}{2} - \frac{1}{2(z+1)^2} \right] dz dy$$

$$= \int_0^1 \left[\frac{1}{2} z + \frac{1}{2(z+1)} \right]_0^{y^2} dy$$

$$= \int_0^1 \left[\frac{y^2}{2} + \frac{1}{2(y^2+1)} - \frac{1}{2} \right] dy$$

$$= \left[\frac{y^3}{6} + \frac{1}{2} \tan^{-1} y - \frac{1}{2} y \right]_0^1$$

$$= \frac{1}{6} + \frac{1}{8} - \frac{1}{2} - 0 = \boxed{\frac{\pi}{8} - \frac{1}{3}}$$

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Q No 3

$$\int_C y \, dx + z \, dy + x \, dz$$

$$C = C_1 + C_2$$

$$C_1: (2, 0, 0) \text{ to } (3, 4, 5)$$

$$C_2: (3, 4, 5) \text{ to } (3, 4, 0)$$

For C_1 : By parametric eqs $\therefore x = x_0 + at$

$$r(t) = \langle 2+t, 4t, 5t \rangle \text{ for } 0 \leq t \leq 1$$

$$dx = dt, \, dy = 4dt, \, dz = 5dt$$

$$\int_0^1 (4t) \, dt + 5t (4dt) + (2+t) (5dt)$$

$$\int_0^1 (4t + 20t + 10 + 5t) \, dt = \int_0^1 (29t + 10) \, dt$$

$$\left| \frac{29t^2}{2} + 10t \right|_0^1 = \frac{29}{2} + 10 = \frac{49}{2}$$

For C_2 : By parametric eqn $\therefore x = x_0 + at$ $(3, 4, 5) \rightarrow (3, 4, 0)$

$$r(t) = \langle 3, 4, 5-5t \rangle \quad 0 \leq t \leq 1$$

$$dx = 0, \, dy = 0, \, dz = -5dt$$

$$\int_0^1 4(0) + (5-5t)(0) + 3(-5dt)$$

$$\int_0^1 -15 \, dt = - \left| 15t \right|_0^1 = -15$$

$$C = C_1 + C_2$$

$$= \frac{49}{2} - 15$$

$$= \frac{19}{2} \approx 9.5$$

(b)

For c,

$$y(t) =$$

$$ds = \sqrt{\left(\frac{dr}{dx}\right)^2}$$

90,

$$= \frac{1}{4} \int_1$$

$$= \frac{1}{6} (5^{3/2} - 1^{3/2}) = \frac{1}{6} (5\sqrt{5} - 1)$$

For C_2 :

also.

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$C =$

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Q No 4 $\int_C x^4 dx + xy dy$

(a)

Using Green's theorem:

$$\therefore \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$P = x^4, Q = xy, \quad \frac{\partial Q}{\partial x} = y, \quad \frac{\partial P}{\partial y} = 0$$

$$= \int_0^1 \int_0^{1-x} (y - 0) dy dx$$

limits: $0 \leq y \leq 1-x$
 $0 \leq x \leq 1$

$$= \int_0^1 \left[\frac{y^2}{2} \right]_0^{1-x} dx$$

$$= \int_0^1 \frac{(1-x)^2}{2} dx = \int_0^1 \frac{(1-2x+x^2)}{2} dx$$

$$= \left[\frac{1}{2}x - \frac{x^2}{2} + \frac{x^3}{6} \right]_0^1 = \frac{1}{2} - \frac{1}{2} + \frac{1}{6}$$

$$\boxed{= \frac{1}{6}}$$

(b) $\int_C (3y - e^{\sin x}) dx + (7x + \sqrt{y^4+1}) dy$

$$C \rightarrow \text{circle } x^2 + y^2 = 9, \quad r=3 \Rightarrow A = \pi r^2 = 9\pi$$

Using Green's theorem:

$$P = 3y - e^{\sin x}, \quad Q = 7x + \sqrt{y^4+1}$$

$$\frac{\partial P}{\partial y} = 3, \quad \frac{\partial Q}{\partial x} = 7$$

$$= \iint_D (7-3) dA = 4 \iint_D dA = 4 \times 9\pi = 36\pi$$

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Q No 5

(a) $\text{Curl } F = \nabla \times F$

$\text{div } F = \nabla \cdot F$

$F(x, y, z) = xz\mathbf{i} + xy\mathbf{j} - y^2\mathbf{k}$

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$$

$$\nabla = \langle z, x, -2y \rangle$$

$$\text{Curl } F = \nabla \times F$$

$\mathbf{i}$	$\mathbf{j}$	$\mathbf{k}$
$\frac{\partial}{\partial x}$	$\frac{\partial}{\partial y}$	$\frac{\partial}{\partial z}$
z	x	$-2y$

$$= \mathbf{i}(-2y-0) - \mathbf{j}(0-x) + \mathbf{k}(y-0)$$

$$= -2y\mathbf{i} + x\mathbf{j} + y\mathbf{k}$$

$$\text{Curl } F = \langle -2y, x, y \rangle$$

$$\text{div } F = \frac{\partial}{\partial x}(xz) + \frac{\partial}{\partial y}(xy)$$

$$+ \frac{\partial}{\partial z}(-y^2)$$

$$= z + x + 0$$

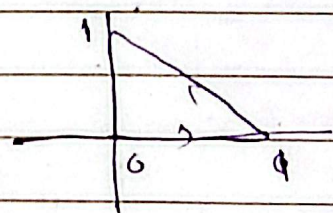
$$\text{div } F = \boxed{x + z}$$

(b) $\int_C (xy \, dx + x^2 \, dy)$

Using Greens Theorem

$$P = xy, \quad Q = x^2$$

$$\frac{\partial P}{\partial y} = x, \quad \frac{\partial Q}{\partial x} = 2x$$



$$= \int_0^1 \int_0^{1-x} (2x - x) \, dy \, dx$$

$$0 \leq y \leq 1-x$$

$$0 \leq x \leq 1$$

$$= \int_0^1 \left[xy \right]_0^{1-x} dx$$

$$= \int_0^1 x - x^2 \, dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \boxed{\frac{1}{6}}$$